

Adaptive Restart Particle Swarm Optimization Based on Search Resource Reallocation

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Abstract—Particle swarm optimization (PSO) is widely used for continuous numerical optimization because of its simple structure and few control parameters. However, standard PSO often suffers from premature convergence when particles rapidly aggregate around local optima and continue consuming function evaluations with limited search contribution. To address this problem, this paper proposes an adaptive restart PSO algorithm based on search resource reallocation, named ARPSO-SRR. The proposed method detects stagnation and diversity loss, adaptively determines restart intensity, and combines global random regeneration with local perturbation around promising regions. The key idea is to reallocate search resources from stagnated or redundant particles to new search positions under the same function evaluation budget. An inefficient particle identification based variant is also investigated as an ablation variant to examine whether a more complex particle selection rule provides additional benefit. Experiments are conducted on 29 functions from the CEC2017 benchmark suite with 30 independent runs. The results show that ARPSO-SRR clearly improves standard PSO and several PSO variants, while remaining lightweight in runtime. Although differential evolution and comprehensive learning PSO achieve stronger overall ranks, ARPSO-SRR provides a simple and effective restart-based enhancement for the PSO framework.

Index Terms—particle swarm optimization, adaptive restart, hybrid restart, search resource reallocation, numerical optimization

I. INTRODUCTION

Particle swarm optimization (PSO) is a population-based metaheuristic inspired by collective behavior and has been widely used for continuous optimization problems [1]. In PSO, each particle updates its position according to its own historical best solution and the best solution found by the swarm. This simple learning mechanism makes PSO easy to implement and suitable for many black-box optimization problems where gradient information is unavailable.

Despite these advantages, standard PSO may suffer from premature convergence. When particles rapidly gather around a local optimum, population diversity decreases and many particles repeatedly search similar positions. These particles still consume function evaluations, but their contribution to further improvement becomes limited. Under a fixed evaluation budget, this phenomenon can be understood as inefficient use of search resources.

Many PSO variants have been proposed to improve exploration and exploitation. Adaptive inertia weight strate-

gies adjust the search tendency during different stages [2]. Stability-oriented and constriction-based analyses improve the understanding of PSO dynamics [3]. Comprehensive learning PSO (CLPSO) maintains diversity by allowing particles to learn from different exemplars [4]. Restart strategies provide another practical solution by regenerating part of the swarm when stagnation occurs.

However, two issues remain important for restart-based PSO. First, restart intensity should not be fixed. A weak restart may be insufficient to escape a local optimum, while an overly strong restart may destroy useful exploitation around promising regions. Second, restart should be interpreted not only as random regeneration, but also as search resource reallocation. In later iterations, some particles provide little new information. Reassigning these particles to new positions can improve the use of the same computational budget.

Based on this motivation, this paper proposes an adaptive restart PSO algorithm based on search resource reallocation, denoted as ARPSO-SRR. The proposed method detects stagnation and diversity loss, determines the restart intensity adaptively, and uses a hybrid restart strategy that combines global random regeneration and local perturbation around the current best region. The proposed method keeps the main structure of PSO unchanged and introduces only lightweight additional operations.

The main contributions of this paper are summarized as follows:

- An adaptive restart PSO framework based on search resource reallocation is proposed to alleviate premature convergence under a fixed function evaluation budget.
- A hybrid restart strategy is designed by combining global random regeneration and local perturbation, so that exploration and exploitation can be balanced during restart.
- An inefficient particle identification based variant is studied as an ablation and comparison variant, which helps analyze whether more complex particle selection brings stable additional improvement.
- Experiments on 29 CEC2017 benchmark functions are conducted. Average ranking, function-group ranking, convergence behavior, restart behavior, runtime analysis, Friedman test, and Wilcoxon signed-rank test are used to evaluate the proposed method.

The remainder of this paper is organized as follows. Section II reviews related work. Section III introduces the proposed ARPSO-SRR. Section IV describes the experimental setup. Section V reports and discusses the experimental results. Section VI concludes this paper.

II. RELATED WORK

A. Standard PSO

Let x_i^t and v_i^t denote the position and velocity of particle i at iteration t . The velocity and position update equations of standard PSO are

$$v_i^{t+1} = wv_i^t + c_1r_1(p_i^t - x_i^t) + c_2r_2(g^t - x_i^t), \quad (1)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1}, \quad (2)$$

where w is the inertia weight, c_1 and c_2 are acceleration coefficients, r_1 and r_2 are uniformly sampled random numbers, p_i^t is the personal best position, and g^t is the global best position.

The performance of PSO depends heavily on the balance between exploration and exploitation. A larger inertia weight usually encourages global exploration, while a smaller inertia weight encourages local exploitation. However, once the swarm is highly concentrated, changing velocity parameters alone may not be sufficient to recover useful diversity.

B. Adaptive and Learning-based PSO Variants

Adaptive PSO variants attempt to improve the search process by changing inertia weights or learning coefficients during evolution [2]. HPSO-TVAC adjusts acceleration coefficients over time to improve the transition from exploration to exploitation. CLPSO allows each dimension of a particle to learn from different exemplars and therefore provides stronger diversity preservation than standard global-best PSO [4]. These methods are useful baselines because they represent parameter adaptation and learning-structure modification.

Compared with these methods, restart-based strategies follow a different direction. They do not necessarily modify the basic velocity update rule. Instead, they reactivate the population by relocating part of the swarm when stagnation or diversity loss is detected. This makes restart-based methods simple and easy to combine with existing PSO variants.

C. Restart Strategies

Restart mechanisms are widely used to reduce the risk of premature convergence. A simple restart strategy regenerates some particles randomly when no improvement is observed for a certain number of iterations. More complex strategies may select particles according to their fitness, distance, contribution, or stagnation history.

From the perspective of resource allocation, a restarted particle is not merely “reset”. Its future function evaluations are transferred from a low-contribution search trajectory to a new candidate region. Therefore, a good restart strategy should answer three questions: when to restart, how many particles to restart, and how to regenerate restarted particles. ARPSO-SRR is designed around these three questions.

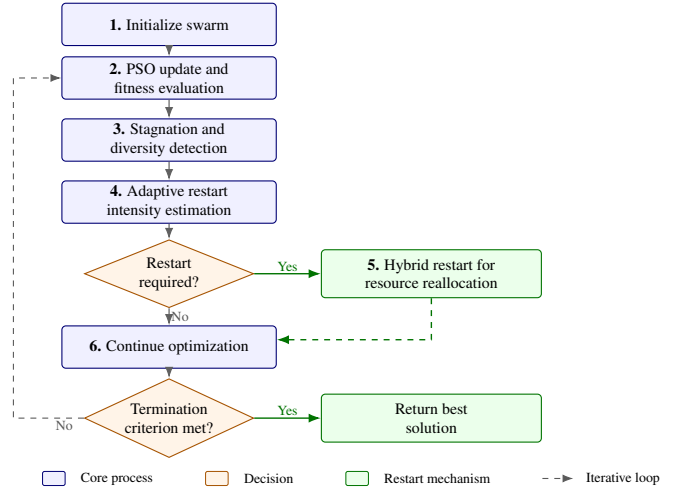


Fig. 1. Framework of the proposed ARPSO-SRR.

III. PROPOSED METHOD

A. Motivation and Overall Framework

In standard PSO, all particles continue to consume function evaluations until the termination condition is met. However, after premature convergence occurs, some particles may repeatedly explore redundant positions around the same local region. ARPSO-SRR treats this phenomenon as inefficient search resource usage. The goal is to identify the search state and reallocate part of the population to more useful regions.

The framework of ARPSO-SRR contains three main components. First, stagnation and diversity indicators are used to detect whether the swarm is losing effective search ability. Second, the restart intensity is calculated adaptively according to the search state. Third, a hybrid restart operator is used to combine global exploration and local exploitation. The overall framework is shown in Fig. 1.

B. Stagnation and Diversity Detection

Population diversity is used to describe the distribution of particles in the search space. In this paper, the diversity measure is defined as

$$D^t = \frac{1}{N} \sum_{i=1}^N \|x_i^t - \bar{x}^t\|_2, \quad (3)$$

where N is the population size and $\bar{x}^t$ is the center of the swarm at iteration t . A smaller value of D^t indicates that particles are more concentrated.

Stagnation is measured by the number of consecutive iterations without global best improvement. Let s^t denote the current stagnation length. When s^t exceeds a predefined threshold or when diversity becomes too low, the restart condition is activated. This design allows ARPSO-SRR to respond to both fitness stagnation and diversity loss.

C. Adaptive Restart Intensity

A fixed restart ratio cannot adapt to different search stages. Therefore, ARPSO-SRR uses an adaptive restart ratio:

$$\rho^t = \rho_{\min} + (\rho_{\max} - \rho_{\min})Q^t, \quad (4)$$

where $\rho_{\min}$ and $\rho_{\max}$ are the lower and upper bounds of the restart ratio, and Q^t is a normalized restart demand indicator. When the swarm suffers from longer stagnation or more serious diversity loss, Q^t becomes larger and more particles are selected for restart. The number of restarted particles is calculated as

$$m^t = \lceil \rho^t N \rceil. \quad (5)$$

This mechanism avoids using the same restart strength in all situations. A mild restart is used when the swarm only shows weak stagnation, while a stronger restart is used when the search state indicates serious convergence.

D. Hybrid Restart Strategy

The restart operation consists of two complementary strategies. The first strategy is global random regeneration:

$$x_i^{t+1} = l + (u - l) \odot r, \quad (6)$$

where l and u are the lower and upper bounds of the search space, r is a random vector sampled from a uniform distribution, and $\odot$ denotes element-wise multiplication. This operator introduces new global exploration ability.

The second strategy is local perturbation around the current global best:

$$x_i^{t+1} = g^t + \sigma^t(u - l) \odot z, \quad (7)$$

where g^t is the current global best position, z is a random vector, and σ^t controls the perturbation strength. In this study, σ^t is bounded by $\sigma_{\min}$ and $\sigma_{\max}$ and gradually reduced with the search process. Therefore, local restart is broader in earlier stages and more exploitative in later stages.

The hybrid strategy avoids two extreme cases. Pure global restart may discard useful information near promising regions, while pure local restart may fail to recover enough diversity. By combining both operators, ARPSO-SRR reallocates search resources while preserving useful exploitation.

E. Algorithm Procedure

The pseudo-code of ARPSO-SRR is given in Algorithm 1. The algorithm first performs standard PSO updates. Then it monitors stagnation and diversity. If the restart condition is satisfied, the adaptive restart ratio is calculated and hybrid restart is applied to selected particles. A visual summary is shown in Fig. 2.

F. Complexity Analysis

The position and velocity update of standard PSO has a computational complexity of $O(ND)$ per iteration, where N is the population size and D is the dimension. The diversity calculation in (3) also requires $O(ND)$ operations. The restart operation modifies at most N particles and therefore has a worst-case complexity of $O(ND)$. As a result, ARPSO-SRR

Algorithm 1 ARPSO-SRR

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1: Initialize particle positions  $x_i$  and velocities  $v_i$ 
2: Evaluate particles and initialize  $p_i$  and  $g$ 
3: Initialize stagnation counter and diversity record
4: while termination criterion is not satisfied do
5:   Update velocities using (1)
6:   Update positions using (2)
7:   Evaluate particles and update  $p_i$  and  $g$ 
8:   Measure diversity  $D^t$  using (3)
9:   Update stagnation counter  $s^t$ 
10:  if restart condition is satisfied then
11:    Calculate restart ratio  $\rho^t$  using (4)
12:    Determine restart number  $m^t$  using (5)
13:    Select particles for search resource reallocation
14:    Apply global random regeneration to part of selected particles
15:    Apply local perturbation around  $g$  to remaining selected particles
16:    Reset stagnation-related records
17:  end if
18: end while
19: return best solution  $g$ 

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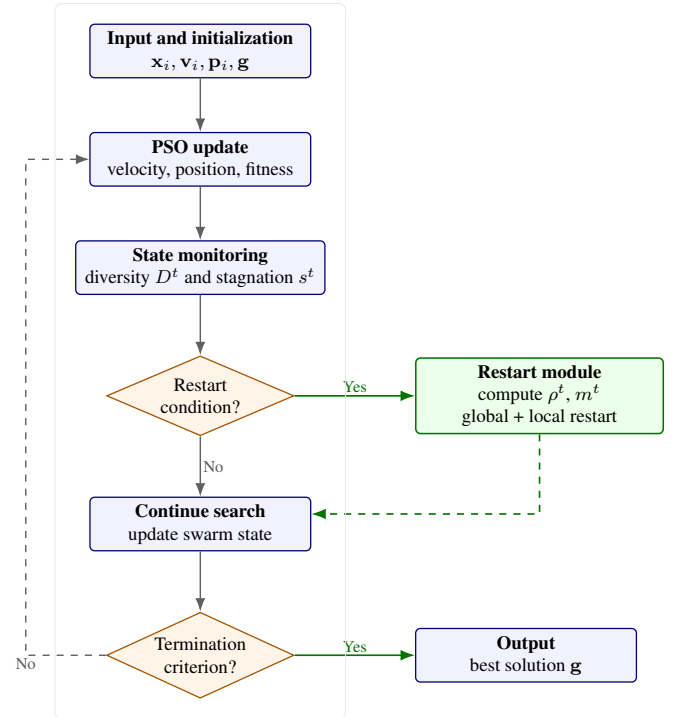


Fig. 2. Visual summary of ARPSO-SRR.

does not change the asymptotic computational complexity of standard PSO. Its additional cost is mainly a small constant factor, and the total number of function evaluations remains unchanged.

TABLE I
CEC2017 FUNCTION GROUPS USED IN THE EXPERIMENTS

Group	Functions
Unimodal	F1, F3
Simple multimodal	F4–F10
Hybrid	F11–F20
Composition	F21–F30

TABLE II
EXPERIMENTAL PARAMETER SETTINGS

Parameter	Setting
Dimension	30
Population size	50
Maximum function evaluations	300000
Independent runs	30
Benchmark functions	CEC2017 F1, F3–F30
Compared algorithms	9
Local perturbation range	$\sigma_{\min} = 0.002, \sigma_{\max} = 0.06$

IV. EXPERIMENTAL SETUP

A. Benchmark Functions

Experiments are conducted on the CEC2017 single-objective real-parameter benchmark suite [5]. In the adopted implementation, F2 is not included, so F1 and F3–F30 are used, resulting in 29 test functions. All experiments are conducted in 30 dimensions. The maximum number of function evaluations is set to $10000D$, namely 300000 evaluations for each run.

The benchmark functions include unimodal, simple multimodal, hybrid, and composition functions. Table I summarizes the function groups used in the analysis.

B. Compared Algorithms and Parameters

ARPSO-SRR is compared with eight algorithms: standard PSO, PSO with adaptive inertia weight (PSO-AW), PSO with random restart (PSO-RS), CLPSO, HPSO-TVAC, genetic algorithm (GA), differential evolution (DE), and ARPSO-EIS. ARPSO-EIS is an inefficient particle identification based variant. It is included mainly as an ablation and comparison variant to examine whether a more explicit particle selection mechanism provides additional improvement over ARPSO-SRR.

All algorithms are executed independently 30 times on each function. The population size is 50 for all algorithms. The same maximum number of function evaluations is used for fair comparison. Table II lists the main experimental settings.

C. Evaluation Metrics

The final optimization result of each run is recorded. Mean value, standard deviation, average rank, runtime, and restart count are used for evaluation. For statistical analysis, the Friedman test is used to compare all algorithms over all functions, and the Wilcoxon signed-rank test is used to compare ARPSO-SRR with each competing algorithm [8]–[10]. A significance level of 0.05 is used.

TABLE III
ABLATION RESULTS OF RESTART-RELATED VARIANTS

Algorithm	Avg. rank	Avg. restarts	Avg. runtime (s)
PSO-RS	6.07	8.94	76.21
ARPSO-Fixed	–	–	–
ARPSO-Global	–	–	–
ARPSO-Local	–	–	–
ARPSO-SRR	4.31	67.72	71.82
ARPSO-EIS	4.45	66.55	72.07

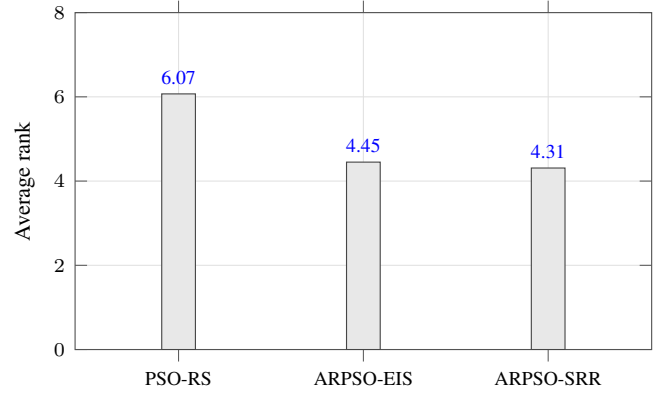


Fig. 3. Average-rank comparison of completed restart-related variants.

V. RESULTS AND DISCUSSION

A. Ablation Study

Table III reports the restart-related variants for which results are already available in the submitted CSV files. The rows of ARPSO-Fixed, ARPSO-Global, and ARPSO-Local are intentionally left blank because the six-variant ablation experiment is still running and should be filled only after the corresponding CSV files are generated.

The completed restart-related results show that ARPSO-SRR obtains a better average rank than PSO-RS and is slightly better than ARPSO-EIS. This suggests that the adaptive hybrid restart framework is more effective than a simple random restart strategy. The comparison with ARPSO-EIS also indicates that the more complex inefficient-particle identification rule does not provide stable additional improvement in the completed experiment.

Fig. 3 is regenerated from the submitted average-rank CSV file. Only the completed restart-related variants are plotted to avoid showing uncompleted ablation variants as numerical results.

B. Overall Performance

Table IV reports the average rank and runtime of all compared algorithms. Lower average rank indicates better overall performance.

DE achieves the best average rank, followed by CLPSO and GA. ARPSO-SRR ranks fourth among the nine compared algorithms. This result should be interpreted carefully. ARPSO-SRR is not designed to replace specialized algorithms

TABLE IV
AVERAGE RANK AND RUNTIME OF COMPARED ALGORITHMS

Algorithm	Avg. rank	Avg. runtime (s)	Relative runtime
DE	1.97	89.88	1.25
CLPSO	3.17	77.40	1.08
GA	4.21	85.44	1.19
ARPSO-SRR	4.31	71.82	1.00
ARPSO-EIS	4.45	72.07	1.00
HPSO-TVAC	5.34	72.82	1.01
PSO-RS	6.07	76.21	1.06
PSO	7.55	74.09	1.03
PSO-AW	7.93	75.37	1.05

such as DE or CLPSO. Its main purpose is to provide a lightweight restart-based enhancement for the PSO framework. Compared with standard PSO, PSO-AW, PSO-RS, and HPSO-TVAC, ARPSO-SRR achieves clearly better average ranking. In addition, its average runtime is 71.82 seconds, which is lower than CLPSO, GA, and DE. This suggests that the proposed restart-based resource reallocation mechanism introduces limited computational overhead.

C. Function-group Analysis

To further analyze the behavior of the compared algorithms, Table V reports the average rank on different function groups.

ARPSO-SRR performs particularly well on composition functions, where its average rank is 3.70. This suggests that adaptive restart and search resource reallocation are useful when the landscape contains multiple interacting structures. On simple multimodal functions, ARPSO-SRR is less competitive than CLPSO and HPSO-TVAC. This may indicate that exemplar-based learning or time-varying acceleration can be more suitable for some regular multimodal landscapes. On hybrid functions, ARPSO-SRR remains competitive with HPSO-TVAC but is weaker than DE, GA, and CLPSO.

Overall, the function-group results support the intended role of ARPSO-SRR: it improves PSO-family methods in a stable and lightweight manner, but it does not dominate all types of benchmark functions.

D. Convergence and Restart Behavior

Representative convergence curves are generated from the submitted mean-curve CSV file. Fig. 4 shows the mean error curves on CEC2017-F23. This function is selected as a representative composition function because it reflects the ability of an algorithm to handle multiple interacting structures. The curve indicates that ARPSO-SRR continues improving after the early stage and obtains a lower final mean error than PSO and PSO-RS on this function, although DE still achieves the best final error.

Restart behavior is generated from the submitted restart-detail CSV file. As shown in Fig. 5, the restart ratio of ARPSO-SRR on CEC2017-F23 becomes active mainly in the middle and later search stages. This is consistent with the design goal of reallocating search resources when stagnation and diversity loss become more likely. Across the completed

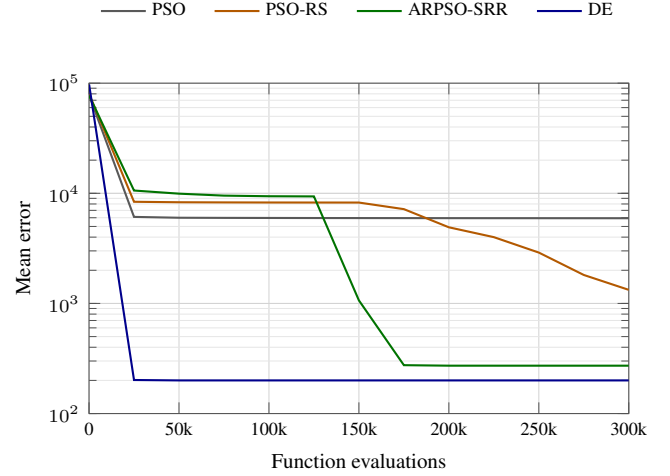


Fig. 4. Representative convergence behavior on CEC2017-F23.

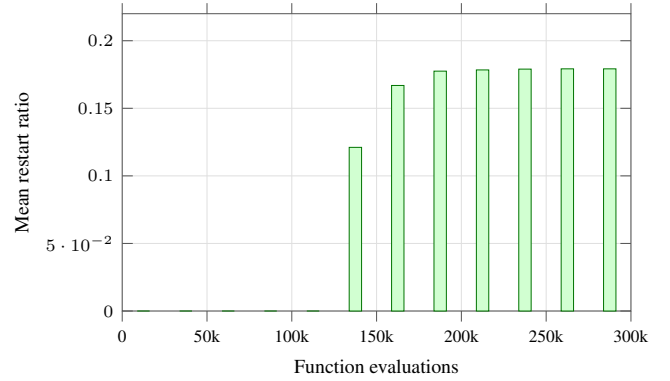


Fig. 5. Mean restart ratio of ARPSO-SRR on CEC2017-F23.

29-function experiment, ARPSO-SRR has an average restart count of 67.72, while PSO-RS has an average restart count of 8.94. These values indicate that ARPSO-SRR performs more active search resource reallocation under the same maximum number of function evaluations.

E. Statistical Analysis

The Friedman test is first conducted over the 29 benchmark functions. The test statistic is 117.2414 and the p -value is 1.2296×10^{-21} , indicating that there are significant differences among the compared algorithms.

Table VII reports the Wilcoxon signed-rank test summary between ARPSO-SRR and each compared algorithm. “Win”, “NSD”, and “Loss” denote the numbers of functions where ARPSO-SRR is significantly better, not significantly different, and significantly worse than the compared algorithm, respectively.

The Wilcoxon results show that ARPSO-SRR significantly outperforms standard PSO on 17 functions, PSO-AW on 19 functions, and PSO-RS on 17 functions. It also obtains more wins than losses against HPSO-TVAC. Compared with

TABLE V
AVERAGE RANK ON DIFFERENT CEC2017 FUNCTION GROUPS

Group	PSO	PSO-AW	PSO-RS	HPSO-TVAC	CLPSO	GA	DE	ARPSO-EIS	ARPSO-SRR
Unimodal	8.00	9.00	5.00	6.50	2.50	6.50	1.50	3.00	3.00
Simple multimodal	7.14	6.29	6.71	4.00	2.57	6.00	3.43	4.00	4.86
Hybrid	7.80	8.20	6.10	4.90	3.40	3.30	1.10	5.40	4.80
Composition	7.50	8.60	5.80	6.50	3.50	3.40	1.90	4.10	3.70

TABLE VI
FRIEDMAN TEST OVER ALL COMPARED ALGORITHMS

Statistic	p -value	Functions	Algorithms
117.2414	1.2296×10^{-21}	29	9

TABLE VII
WILCOXON SIGNED-RANK TEST SUMMARY FOR ARPSO-SRR

Compared algorithm	Win	NSD	Loss
PSO	17	11	1
PSO-AW	19	10	0
PSO-RS	17	11	1
HPSO-TVAC	13	9	7
GA	11	9	9
ARPSO-EIS	1	27	1
CLPSO	7	6	16
DE	3	3	23

GA, ARPSO-SRR is relatively balanced, with 11 wins, 9 nonsignificant differences, and 9 losses. However, ARPSO-SRR is weaker than CLPSO and DE in overall statistical comparison. This result is consistent with the average rank results in Table IV.

The comparison between ARPSO-SRR and ARPSO-EIS is also important. ARPSO-SRR has 1 win, 27 nonsignificant differences, and 1 loss against ARPSO-EIS. Therefore, the EIS-based selection rule does not provide stable additional improvement. This supports using ARPSO-SRR as the main method and treating ARPSO-EIS as an ablation variant.

F. Discussion

The experimental results indicate that ARPSO-SRR is suitable for improving PSO-family algorithms. Its main advantage is not that it beats every strong evolutionary algorithm. Instead, it provides a simple, lightweight, and easy-to-implement mechanism for reducing premature convergence in PSO.

Compared with PSO, PSO-AW, and PSO-RS, ARPSO-SRR achieves clear improvement. This suggests that resource reallocation through adaptive hybrid restart is more effective than only adjusting inertia weight or using simple random restart. Compared with HPSO-TVAC, ARPSO-SRR also shows competitive performance. Against GA, the results are balanced. Against CLPSO and DE, ARPSO-SRR is weaker in average rank and statistical comparison, which indicates that more specialized learning and differential mutation mechanisms still have strong advantages on many CEC2017 functions.

The function-group analysis provides a more detailed view. ARPSO-SRR performs well on composition functions, suggesting that restart-based resource reallocation can help when the search landscape contains multiple complex structures. However, on some simple multimodal and hybrid functions, ARPSO-SRR is not always superior. This may be because frequent restart may interrupt long-term exploitation when the landscape requires continuous refinement rather than diversity recovery. This limitation can be addressed in future work by designing more adaptive restart triggers and restart intensity control.

VI. CONCLUSION

This paper proposed ARPSO-SRR, an adaptive restart PSO algorithm based on search resource reallocation. The proposed method detects stagnation and diversity loss, adaptively determines restart intensity, and combines global random regeneration with local perturbation around promising regions. The method keeps the original PSO framework simple and does not increase the maximum number of function evaluations.

Experiments on 29 CEC2017 benchmark functions demonstrate that ARPSO-SRR improves standard PSO and several PSO variants. The Friedman test confirms significant overall differences among compared algorithms, and Wilcoxon tests show that ARPSO-SRR clearly outperforms PSO, PSO-AW, and PSO-RS on many functions. The comparison with ARPSO-EIS indicates that more complex inefficient particle identification does not necessarily bring stable additional performance gains. Although DE and CLPSO achieve stronger overall performance, ARPSO-SRR provides a lightweight and effective restart-based enhancement for PSO. Future work will investigate more adaptive restart triggers, large-scale optimization problems, and real-world engineering applications.

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